OWA Portfolio Optimization: a Disciplined Convex Programming Framework

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[First version: November 2021] [Latest version: May 2022]

Abstract

This work presents the ordered weighted average or OWA portfolio optimization model. This general model allows us to express various classical portfolio optimization models like Gini mean difference, conditional value at risk, weighted conditional value at risk, tail Gini and minimax as functions of ordered statistics. Also, we propose a new group of risk measures that we called range risk measures that can be optimized using the OWA portfolio optimization model. Then, we run some numerical examples of the OWA portfolio optimization framework using Python, CVXPY and MOSEK solver. Finally, we test the efficiency of the new framework in medium scale problems respect to classical formulations.

Keywords: risk analysis, portfolio optimization, ordered weighted average, portfolio selection, gini mean difference, risk parity.

JEL Codes: C61, G11

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1 Introduction

The ordered weighted average (OWA) operator was introduced by Yager (1988). This operator aggregates the values assigning the weights to the ordered values and allows to model various aggregation preferences. Also, Yager (1996) show that for arbitrary weights, this problem can be posed as a Mixed Integer Linear Programming (MILP) problem, however this formulation is computationally intensive. On the other hand, Ogryczak and Śliwiński (2003) showed that in the case on monotonic weights, this case can be posed as a Linear Programming (LP) problem and Chassein and Goerigk (2015) proposed a most efficient way to solve this case problem using combinatorial optimization techniques.

In the context of portfolio optimization there are few studies that apply the OWA operator in the portfolio optimization process. Ogryczak (2000) proposed a multiple criteria linear programming model for portfolio optimization that allows to model the preferences of investors, however this model is not efficient in medium or large scale problems due to the number of constraints are proportional to a factorial of the number of observations. Laengle et al. (2017) showed how the OWA operator can model different degrees of optimism and pessimism of investors in the context of mean variance portfolio problem.

Convex programming is a subclass of nonlinear programming (NLP) that unifies and generalizes least squares (LS), linear programming (LP), convex quadratic programming (QP) and cone programming (CP). Disciplined convex programming, defined in Liberti (2006), is a modeling methodology that imposes a set of conventions that one must follow when constructing convex programs. The advantage of this methodology is that the transformations necessary to convert disciplined convex programs into solvable form can be fully automated. There are several optimization packages that supports disciplined convex programs, for example CVXPY, CVX, MOSEK, among other packages.

This work proposes the OWA portfolio optimization model, a general portfolio model that group various classical portfolio optimization models like Gini mean difference (GMD) proposed by Yitzhaki (1982), conditional value at risk (CVaR) proposed by Rockafellar and Uryasev (2000), weighted CVaR (WCVaR) and tail Gini proposed by Ogryczak and Ruszczynskia (2002), and minimax proposed by Young (1998). We also show how we can build a new group of risk measures that we called *range risk measures* as special cases of OWA operator. First, we show how the GMD can be expressed using an OWA operator. Then, we show how the OWA optimization model can be combined with five portfolio optimization problems using disciplined convex programming techniques. Then, we show how various classical portfolio models and range risk measures portfolio models are special cases of the OWA portfolio. Finally, we make numerical examples of OWA portfolio optimization using Python and MOSEK solver.

2 Portfolio Optimization Frameworks

2.1 Mean-Risk Portfolio Optimization

The mean-risk optimization is a generalization of the model proposed by Markowitz (1952), that can be applied to any convex risk measure. Mean-risk optimization begins selecting $N \in \mathbb{N}$ assets, a series of historical returns $r \in \mathbb{R}^{T \times N}$ where $T \in \mathbb{N}$ is the number of observations, $\mu \in \mathbb{R}^{N}$ is the vector of expected returns of r, $x \in \mathbb{R}^{N}$ is the vector of weights or portfolio, μx^{τ}[1] is the expected return of portfolio x, $\phi(x)$ is the expected risk of portfolio x and $r x^{\tau}$ is series of historical portfolio returns. The mean-risk optimization can be resumed as four convex optimization problems:

- Maximization of the expected return given a level of risk $\bar{\phi}$.

$$
\begin{aligned}
\max_{x} \quad & \mu x^{\tau} \\
\text{s.t.} \quad & \phi(x) \leq \bar{\phi} \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall \, i = 1, \ldots, N
\end{aligned} \tag{1}
$$

- Minimization of the expected risk given a level of expected return $\bar{\mu}$.

$$
\begin{aligned}
\min_{x} \quad & \phi(x) \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall \, i = 1, \ldots, N
\end{aligned} \tag{2}
$$

- Maximization of a risk averse utility function given a risk aversion parameter

[1] The operator $*^{\tau}$ is the operator transpose of a matrix.

$\lambda \in [0, \infty)$.

$$\begin{aligned} \max_{x} \quad & \mu x^{\tau} - \lambda \phi(x) \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = 1 \\ & x_i \geq 0 \ ; \ \forall \ i = 1, \ldots, N \end{aligned} \tag{3}$$

- Maximization of the risk adjusted return, based on the idea of Sharpe (1964), assuming that the return of the risk free asset is r_f. This problem can be posed as the following form:

$$\begin{aligned} \max_{x} \quad & \frac{\mu x^{\tau} - r_f}{\phi(x)} \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = 1 \\ & x_i \geq 0 \ ; \ \forall \ i = 1, \ldots, N \end{aligned} \tag{4}$$

This problem is quasi convex, so it cannot be solved with a convex programming software; however, it can be transformed into a convex programming problem using the transformation of Charnes and Cooper (1962) as long as it is assumed that $\mu x^{\tau} - r_f > 0$. After the transformation the problem takes the following form:

$$\begin{aligned} \min_{z,\,k} \quad & \phi(z) \\ \text{s.t.} \quad & \mu z^{\tau} - r_f k = 1 \\ & \sum_{i=1}^{N} z_i = k \\ & k \geq 0 \\ & z_i \geq 0 \ ; \ \forall \ i = 1, \ldots, N \end{aligned} \tag{5}$$

where the optimal portfolio is obtained making the transformation $x = z/k$.

2.2 Risk Parity Portfolio Optimization

The general vanilla risk parity portfolio formulated by Bruder and Roncalli (2012) has the following form:

$$\begin{aligned}
\min_{z} \quad & \phi(z) \\
& \sum_{i=1}^{N} b_i \ln(z_i) \geq c \\
& \sum_{i=1}^{N} b_i = 1 \\
& z_i \geq 0 \ ; \ \forall \ i = 1, \ldots, N
\end{aligned} \tag{6}$$

Where b is a vector of risk concentration constraints of size $1 \times N$ and c is an arbitrary constant. The optimal portfolio is obtained making the transformation $x = z / \sum_{i=1}^{N} z_i$.

3 Gini Mean Difference

3.1 Linear Programming Models

GMD portfolio optimization appears in academic literature like Mansini et al. (2003), as the following problem:

$$\begin{aligned}
\min_{x} \quad & \frac{1}{T(T-1)} \sum_{i=1}^{T} \sum_{j \neq i}^{T} |r_i x - r_j x| \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \ ; \ \forall \ i = 1, \ldots, N
\end{aligned} \tag{7}$$

This problem can be posed as a linear programming problem as in Mansini et al. (2015) using the auxiliary variables d as follows:

$$
\begin{aligned}
\min_{x,d} \quad & \frac{1}{T(T-1)} \sum_{i=1}^{T} \sum_{j>i}^{T} d_{i,j} \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& d_{i,j} \geq (r_i - r_j) x^{\tau} \; ; \; \forall\, i,j = 1,\ldots,T \; ; \; i \neq j \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall\, i = 1,\ldots,N \\
& d_{i,j} \geq 0 \; ; \; \forall\, i,j = 1,\ldots,T
\end{aligned} \tag{8}
$$

However, the original model proposed by Yitzhaki (1982), taking advantage that $|r_i x - r_j x| = |r_j x - r_i x|$ and $|r_i x - r_i x| = 0$, proposed a most compressed formulation:

$$
\begin{aligned}
\min_{x} \quad & \frac{2}{T(T-1)} \sum_{i=1}^{T} \sum_{j>i}^{T} |r_i x - r_j x| \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall\, i = 1,\ldots,N
\end{aligned} \tag{9}
$$

This most compressed model can be posed in a linear programming model of the form:

$$
\begin{aligned}
\min_{x,d} \quad & \frac{2}{T(T-1)} \sum_{i=1}^{T} \sum_{j>i}^{T} d_{i,j} \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& d_{i,j} \geq (r_i - r_j) x^{\tau} \; ; \; \forall\, i > j = 1,\ldots,T \\
& d_{i,j} \geq -(r_i - r_j) x^{\tau} \; ; \; \forall\, i > j = 1,\ldots,T \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0 \; ; \; \forall\, i = 1,\ldots,N
\end{aligned} \tag{10}
$$

we drop the constraint $d_{i,j} \geq 0$ because is redundant and increase the computation time. Finally, both problems are very difficult to solve because the number of variables are proportional to $T(T-1)$ and $T(T-1)/2$ respectively.

3.2 L-Moments

L-moments or linear moments are expectations of certain linear combination of order statistics. As shown in Hosking (1990), L-moments are used to summarize the shape of a probability distribution in a similar way than central moments.

If $\{y_{[1]} < \cdots < y_{[i]} < \cdots < y_{[n]}\}$ is the ordered sample, where $y_{[i]}$ is the i-ordered statistic. The r-th sample L-moment can be calculated using the following formula:

$$\lambda_r = \binom{T}{r}^{-1} \sum \sum_{1 \leq i_1 < i_2 \cdots < i_r \leq n} \cdots \sum \frac{1}{r} \sum_{k=0}^{r-1} (-1)^k \binom{r-1}{k} y_{[i_{r-k}]} \tag{11}$$

On the other hand, Wang (1996) proposed direct estimators for the first four L-moments in a finite sample of T observations using the following formulas:

$$\begin{aligned}
\lambda_1 &= \binom{T}{1}^{-1} \sum_{i=1}^{T} y_{[i]} \\
\lambda_2 &= \tfrac{1}{2}\binom{T}{2}^{-1} \sum_{i=1}^{T} \left[\binom{i-1}{1} - \binom{T-i}{1}\right] y_{[i]} \\
\lambda_3 &= \tfrac{1}{3}\binom{T}{3}^{-1} \sum_{i=1}^{T} \left[\binom{i-1}{2} - 2\binom{i-1}{1}\binom{T-i}{1} + \binom{T-i}{2}\right] y_{[i]} \\
\lambda_4 &= \tfrac{1}{4}\binom{T}{4}^{-1} \sum_{i=1}^{T} \left[\binom{i-1}{3} - 3\binom{i-1}{2}\binom{T-i}{1} + 3\binom{i-1}{1}\binom{T-i}{2} - \binom{T-i}{3}\right] y_{[i]}
\end{aligned} \tag{12}$$

As shown in Hosking (1990), the GMD is the double of the λ_2, so the GMD can be calculated as follows:

$$\begin{aligned}
\text{GMD} &= 2\lambda_2 \\
\text{GMD} &= \binom{T}{2}^{-1} \sum_{i=1}^{T} \left\{\binom{i-1}{1} - \binom{T-i}{1}\right\} y_{[i]} \\
\text{GMD} &= \sum_{i=1}^{T} \left(2\left[\frac{2i-1-T}{T(T-1)}\right]\right) y_{[i]}
\end{aligned} \tag{13}$$

This formulation allows us to express the GMD using an OWA operator.

4 OWA Portfolio Optimization

4.1 OWA Optimization

The ordered weighted averaging operator or OWA operator was proposed by Yager (1988) and it is defined as $\sum_i^T w_i y_{[i]}$ where $y_{[i]}$ is the i-ordered statistic. This operator can be used in optimization problems using the MILP formulation proposed by Yager (1996). In the case of monotonic[2] weights, Ogryczak and Śliwiński (2003) shown than the OWA operator can be modeled as a linear programming problem of the form:

$$\begin{aligned} \min_x \quad & \sum_{j=1}^{T} j w'_j r_j - \sum_{j=1}^{T}\sum_{j=1}^{T} j w'_j d_{ij} \\ \text{s.t.} \quad & Cx^\tau = y \\ & d_{ij} \geq r_j - y_{[i]} \; \forall \, i, j = 1, \ldots, T \\ & x \in \mathcal{X} \end{aligned} \tag{14}$$

where $w'_j = w_j - w_{j+1}$, $w'_k = w_k$, and r and d are auxiliary variables.

However, this formulation is very intensive computationally because it requires $T^2 + 2T$ additional variables and $T^2 + T$ new constraints. On the other hand, Chassein and Goerigk (2015) proposed a most efficient way to solve the OWA optimization problem taking advantage that the problem can be posed as a combinatorial optimization problem:

$$\begin{aligned} \max_x \quad & \sum_{i=1}^{T} \alpha_i + \beta_i \\ \text{s.t.} \quad & Cx^\tau = y \\ & \alpha_i + \beta_j \leq w_i y_{[j]} \; \forall \, i, j = 1, \ldots, T \\ & x \in \mathcal{X} \end{aligned} \tag{15}$$

where α and β are auxiliary variables. This formulation is more efficient computationally because it requires only $3T$ additional variables and $T^2 + T$ new constraints.

[2]Nonincreasing weights are of the form $w_1 \geq w_2 \geq \ldots \geq w_{T-1} \geq w_T$ and nondecreasing weights are of the form $w_1 \leq w_2 \leq \ldots \leq w_{T-1} \leq w_T$.

4.2 OWA Portfolio Optimization Models

Combining the OWA optimization model with the five portfolio optimization problems from section 2, we obtain the following problems:

- Maximization of the expected return given a level of OWA risk $\overline{\phi}_{\text{OWA}}$.

$$
\begin{aligned}
\max_{\alpha,\beta,x,y} \quad & \mu x^{\tau} \\
\text{s.t.} \quad & \sum_{i=1}^{T} \alpha_i + \beta_i \leq \overline{\phi}_{\text{OWA}} \\
& r x^{\tau} = y \\
& \alpha_i + \beta_j \geq w_i y_{[j]} \quad \forall\, i, j = 1, \dots, T \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0\ ;\ \forall\, i = 1, \dots, N
\end{aligned}
\tag{16}
$$

  where r_j is a vector of returns in period j for the N assets and x is the vector of weights of the assets.

- Minimization of the expected OWA risk given a level of expected return $\bar{\mu}$.

$$
\begin{aligned}
\min_{\alpha,\beta,x,y} \quad & \sum_{i=1}^{T} \alpha_i + \beta_i \\
\text{s.t.} \quad & \mu x^{\tau} \geq \bar{\mu} \\
& r x^{\tau} = y \\
& \alpha_i + \beta_j \geq w_i y_{[j]} \quad \forall\, i, j = 1, \dots, T \\
& \sum_{i=1}^{N} x_i = 1 \\
& x_i \geq 0\ ;\ \forall\, i = 1, \dots, N
\end{aligned}
\tag{17}
$$

- Maximization of OWA risk averse utility function given a risk aversion parameter

$\lambda \in [0, \infty)$:

$$
\begin{aligned}
\max_{\alpha, \beta, x, y} \quad & \mu x^{\tau} - \lambda \left(\sum_{i=1}^{T} \alpha_i + \beta_i \right) \\
\text{s.t.} \quad & \sum_{i=1}^{N} x_i = 1 \\
& r x^{\tau} = y \\
& \alpha_i + \beta_j \geq w_i y_{[j]} \quad \forall \, i, j = 1, \ldots, T \\
s \quad & \sum_{i=1}^{N} x_i = k \\
& x_i \geq 0 \; ; \; \forall \, i = 1, \ldots, N
\end{aligned}
\tag{18}
$$

- Maximization of the OWA risk adjusted return assuming that the return of the risk free asset is r_f.

$$
\begin{aligned}
\min_{\alpha, \beta, y, z} \quad & \sum_{i=1}^{T} \alpha_i + \beta_i \\
\text{s.t.} \quad & \mu z^{\tau} - r_f k = 1 \\
& r z^{\tau} = y \\
& \alpha_i + \beta_j \geq w_i y_{[j]} \quad \forall \, i, j = 1, \ldots, T \\
& \sum_{i=1}^{N} x_i = k \\
& x_i \geq 0 \; ; \; \forall \, i = 1, \ldots, N
\end{aligned}
\tag{19}
$$

where the optimal portfolio is obtained making the transformation $x = z/k$.

- Risk parity OWA risk portfolio.

$$
\begin{aligned}
\min_{\alpha, \beta, y, z} \quad & \sum_{i=1}^{T} \alpha_i + \beta_i \\
& \sum_{i=1}^{N} b_i \ln(z_i) \geq c \\
& r z^{\tau} = y \\
& \alpha_i + \beta_j \geq w_i y_{[j]} \quad \forall \, i, j = 1, \ldots, T \\
& \sum_{i=1}^{N} b_i = 1 \\
& z_i \geq 0 \; ; \; \forall \, i = 1, \ldots, N
\end{aligned}
\tag{20}
$$

where the optimal portfolio is obtained making the transformation $x = z / \sum_{i=1}^{N} z_i$.

4.3 Some Special Cases of OWA Portfolio Optimization

We can get OWA equivalents of several classical risk measures depending on the values of the vector of weights w:

- We get the GMD model when $w_i^{\mathrm{GMD}} = 2\left(\frac{2i-1-T}{T(T-1)}\right)$.
- We get the worst case of portfolio returns or minimax model when $w_1^{\mathrm{WC}} = -1$ and $w_i^{\mathrm{WC}} = 0 \; \forall i = 2, \ldots, T$.
- We get the CVaR(α) model when $w_i^{\mathrm{CVaR}(\alpha)} = -\frac{1}{\alpha T} \; \forall i = 1, \ldots, k-1$, $w_k = -1 + \sum_{i=1}^{k-1} \frac{1}{\alpha T}$ and $k = \lceil \alpha T \rceil$[3].
- We get the WCVaR model when $w^{\mathrm{WCVaR}(\bar{\alpha}, p)} = \sum_{j=1}^{J} p_j w^{\mathrm{CVaR}(\alpha_j)}$, $\sum_{j=1}^{J} p_j = 1$ and $\bar{\alpha} = [\alpha_1, \ldots, \alpha_J]$.
- We get the tail Gini model when $w^{\mathrm{TG}(\alpha)} = \sum_{j=1}^{m} p_j w^{\mathrm{CVaR}(\alpha_j)}$, where $p_1 = \frac{\alpha_1 \alpha_2}{\alpha^2}$, $p_j = \frac{(\alpha_{i+1} - \alpha_{i-1})\alpha_i}{\alpha^2}$ for $j = 2, \ldots, m-1$, $p_m = \frac{(\alpha_{m-1} - \alpha_m)}{\alpha}$ and a grid of significance levels $0 = \alpha_1 < \ldots < \alpha_j < \ldots < \alpha_m = \alpha$.

[3]Ceiling operator

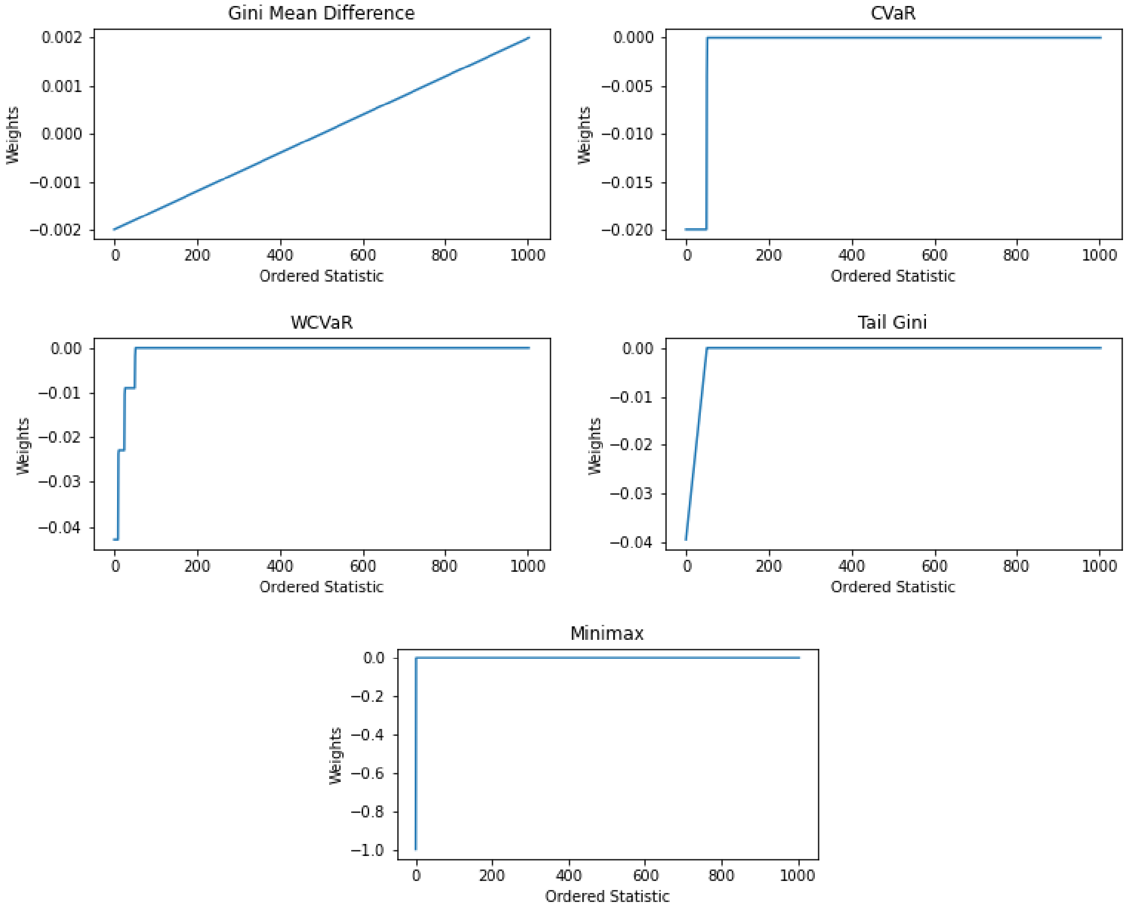

Figure 1: OWA Weights of Classical Risk Measures

In addition, we can define a new group of dispersion risk measures called range risk measures that measure the dispersion of returns based on the distance between of ordered statistics in left and right tail. Some special cases are:

- The range of portfolio returns is the difference between highest and lowest returns. We get this model when $w_1^{\text{Range}} = -1$, $w_i^{\text{Range}} = 0 \, \forall i = 2, \ldots, T-1$ and $w_T^{\text{Range}} = 1$.
- The CVaR range is the difference between CVaR of upper tail and CVaR of lower tail. We get this model when $w^{\text{CVaR-Range}(\alpha,\beta)} = w^{\text{CVaR}(\alpha)} - Aw^{\text{CVaR}(\beta)}$ where A is the reversal matrix and it is defined as:

$$A = \begin{bmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \ldots & 1 & 0 \\ \vdots & \vdots & \iddots & \vdots & \vdots \\ 0 & 1 & \ldots & 0 & 0 \\ 1 & 0 & \ldots & 0 & 0 \end{bmatrix}$$

- The WCVaR range is the difference between WCVaR of upper tail and WCVaR of lower tail. We get this model when $w^{\text{WCVaR-Range}(\bar{\alpha}, p^{\alpha}, \bar{\beta}, p^{\beta})} = w^{\text{WCVaR}(\bar{\alpha}, p^{\alpha})} - Aw^{\text{WCVaR}(\bar{\beta}, p^{\beta})}$ where $\sum_{j=1}^{J} p_j^{\alpha} = 1$, $\sum_{l=1}^{L} p_l^{\beta} = 1$, $\bar{\alpha} = [\alpha_1, \ldots, \alpha_J]$ and $\bar{\beta} = [\beta_1, \ldots, \beta_L]$.

- The tail Gini range is the difference between tail Gini of upper tail and tail Gini of lower tail. We get this model when $w^{\text{TG-Range}(\alpha,\beta)} = w^{\text{TG}(\alpha)} - Aw^{\text{TG}(\beta)}$.

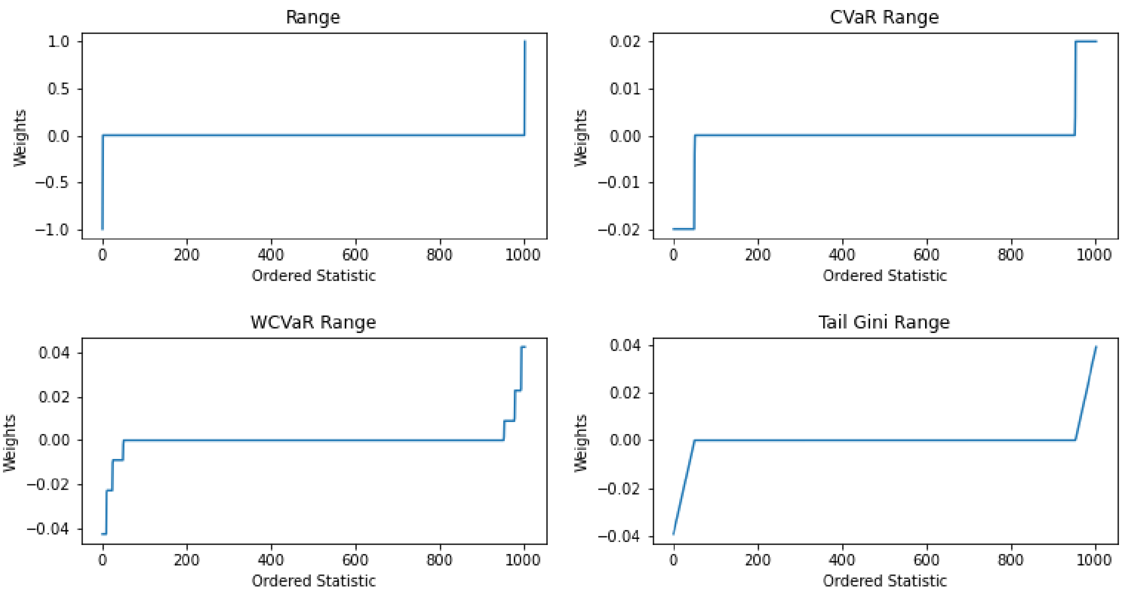

Figure 2: OWA Weights of Range Risk Measures

Additionally, we can define a personalized range risk measure using the formula $w = w^a - Aw^b$ to build weights. Finally, we can define a personalized OWA risk measure building monotonic nondecreasing weights.

5 Numerical Examples

We select 30 assets (i.e., stocks JCI, TGT, CMCSA, CPB, MO, T, APA, MMC, JPM, ZION, PSA, BAX, BMY, AAPL, PCAR, BA, TMO, TXT, DE, MSFT, HPQ, SEE, VZ, CNP, NI, JNJ, PFE, AMZN, GE and GOOG) from the S&P 500 (NYSE) and download daily adjusted closed prices from Yahoo Finance for the period from January 1, 2016 to December 30, 2019. Then, we calculated daily returns building a returns matrix of size $T = 1004$ and $N = 30$. To calculate the portfolios we consider a significance level of 5% and we use Python 3.7[4], CVXPY[5] and MOSEK[6] solver.

5.1 GMD Numerical Example

In this section, we calculate the portfolio that maximize the GMD adjusted return, the GMD efficient frontier and risk parity GMD portfolio.

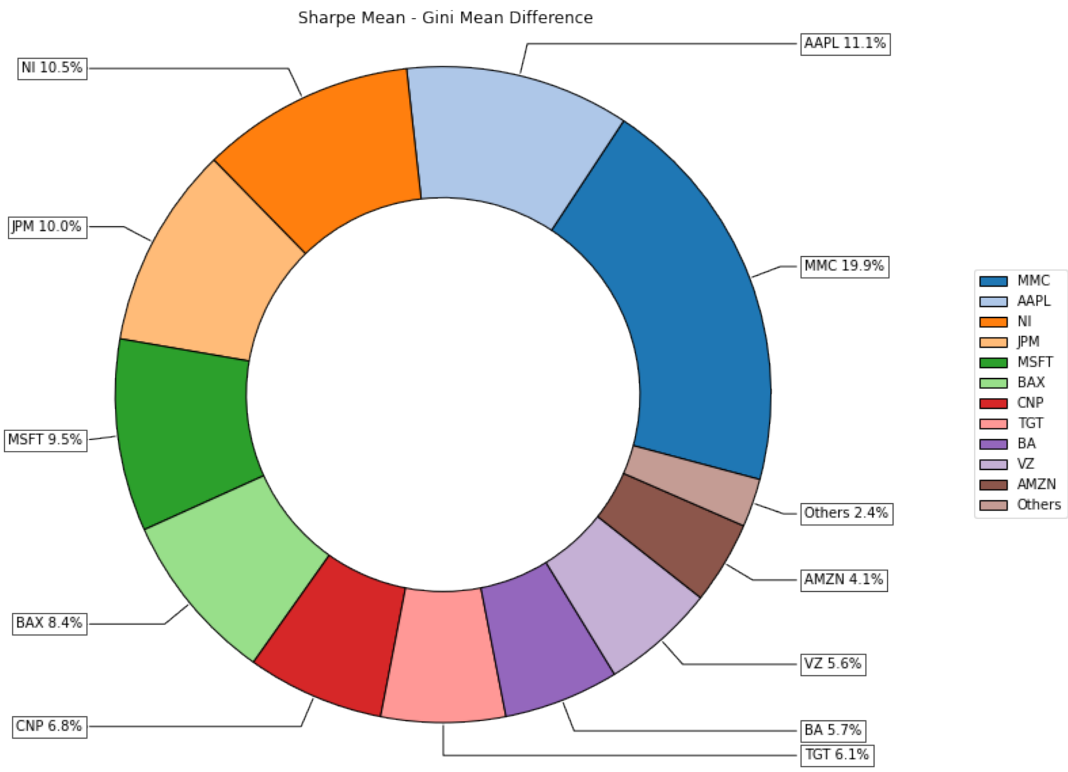

Figure 3: Composition Max GMD Adjusted Return Portfolio

[4]https://winpython.github.io
[5]Diamond and Boyd (2016) and Agrawal et al. (2018)
[6]https://www.mosek.com

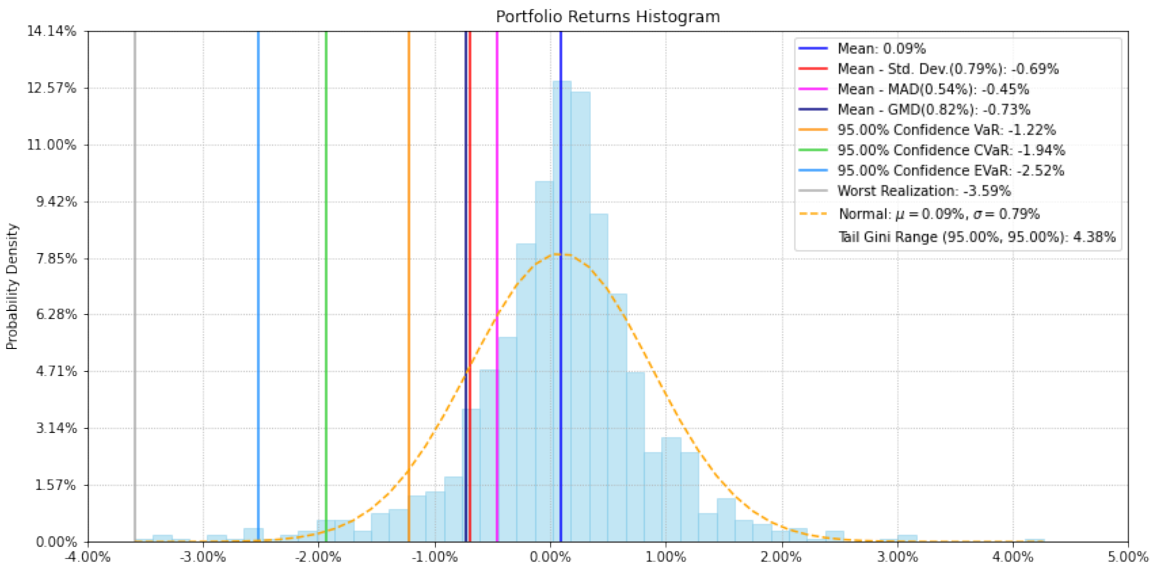

Figure 4: Histogram Max GMD Adjusted Return Portfolio

If we see the histogram of max GMD adjusted return portfolio in Figure 4, the standard deviation is 0.79%, the mean absolute deviation (MAD) is 0.54% and GMD is 0.82%.

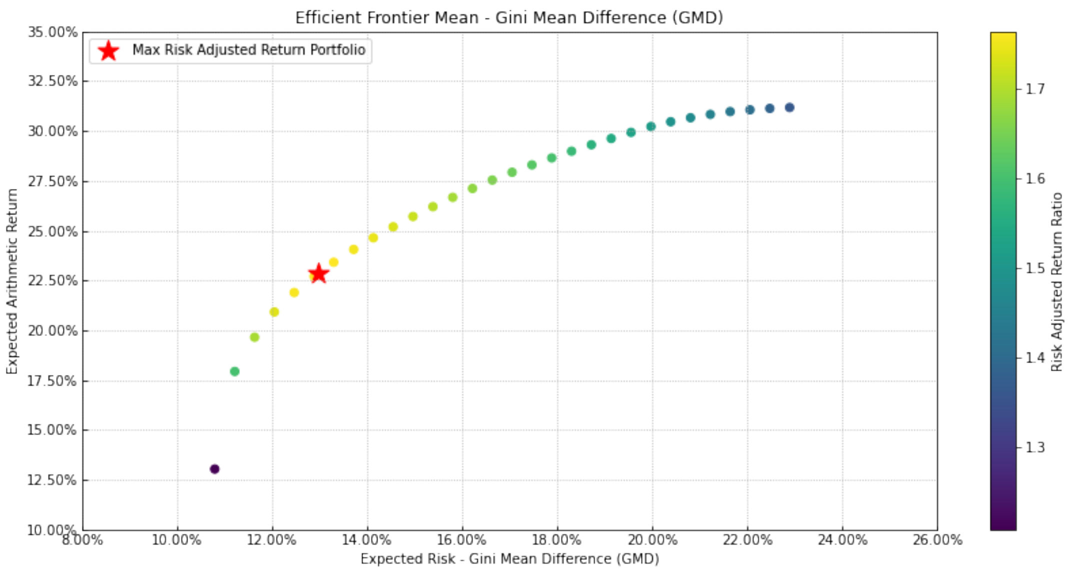

Figure 5: Efficient Frontier Return-GMD

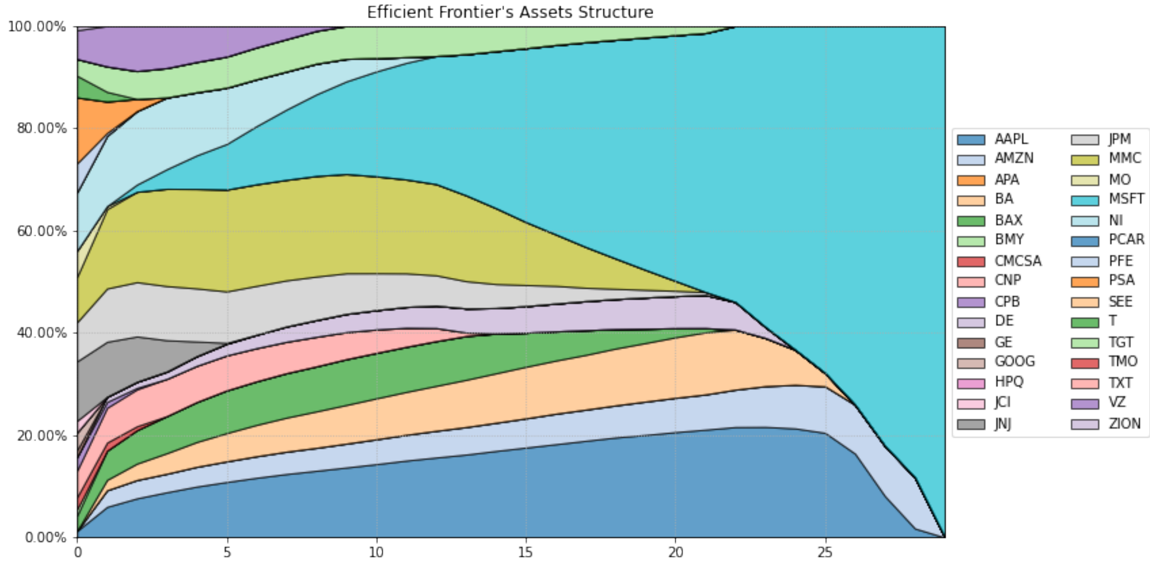

Figure 6: Composition Efficient Frontier Return-GMD

To build the efficient frontier Return-GMD we solve the problem maximize the portfolio return subject to a constraint on GMD for several values of GMD.

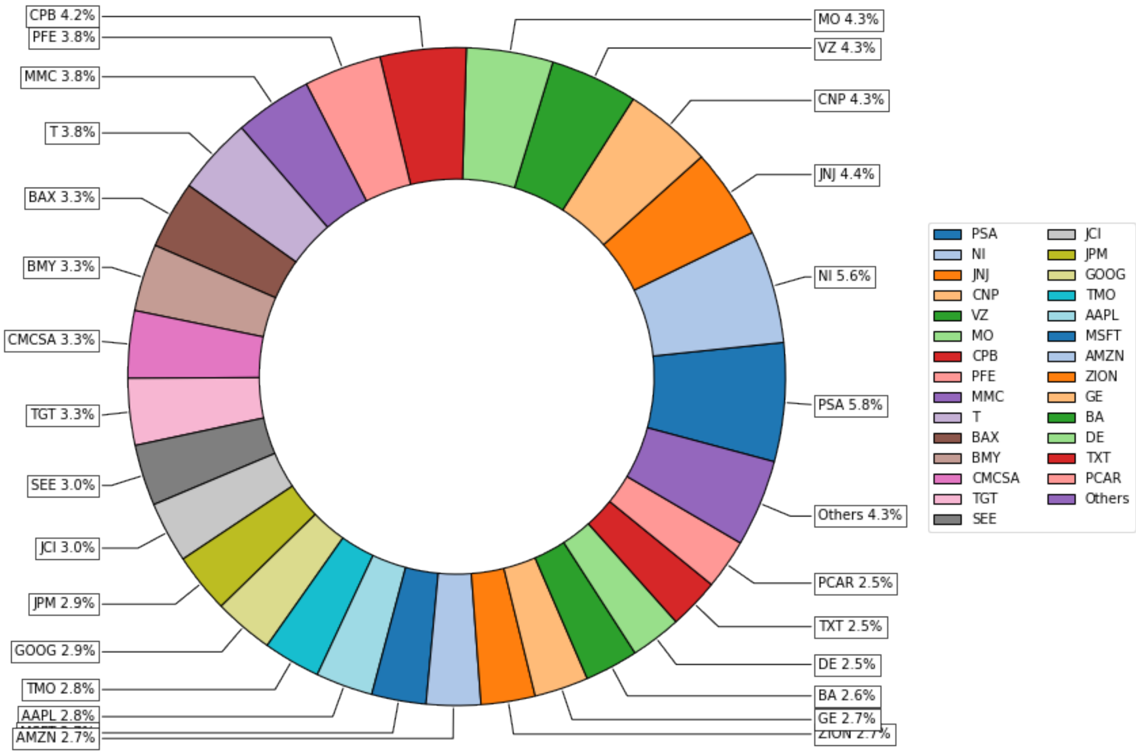

Figure 7: Composition Risk Parity GMD Portfolio

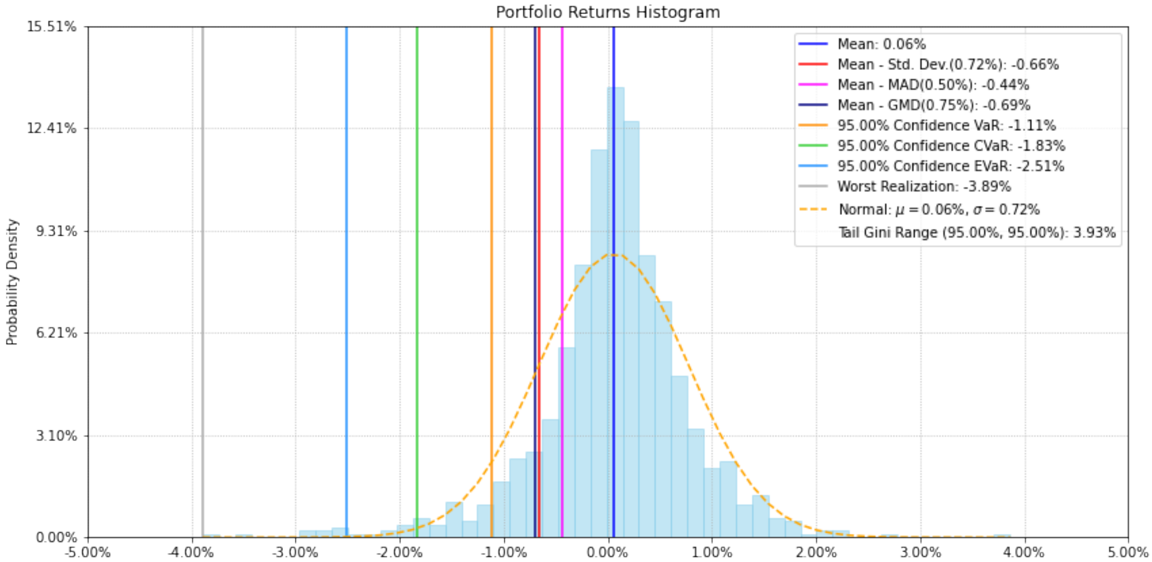

Figure 8: Histogram Risk Parity GMD Portfolio

In the histogram of risk parity GMD portfolio in Figure 8, the standard deviation is 0.72%, MAD is 0.50% and GMD is 0.75%. We can see that these values are lower than the values observed in 4. This behavior has sense because GMD is a deviation risk measure and the risk parity approach tries to make more stable the portfolio returns.

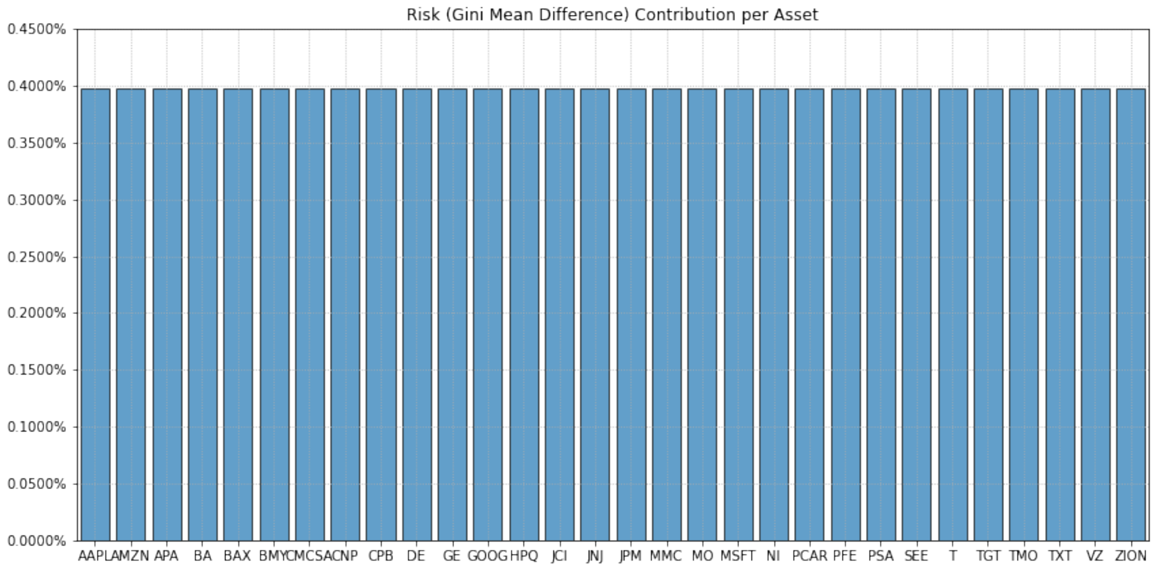

Figure 9: GMD Contribution per Asset

In Figure 9, we can see that risk parity GMD portfolio effectively breaks down GMD evenly across all assets. This shows that the solution of risk parity GMD problem that we posed fulfills its objective.

5.2 Tail Gini Range Numerical Example

In this section, we calculate the portfolio that maximize the tail Gini range adjusted return, the tail Gini range efficient frontier and risk parity tail Gini range portfolio.

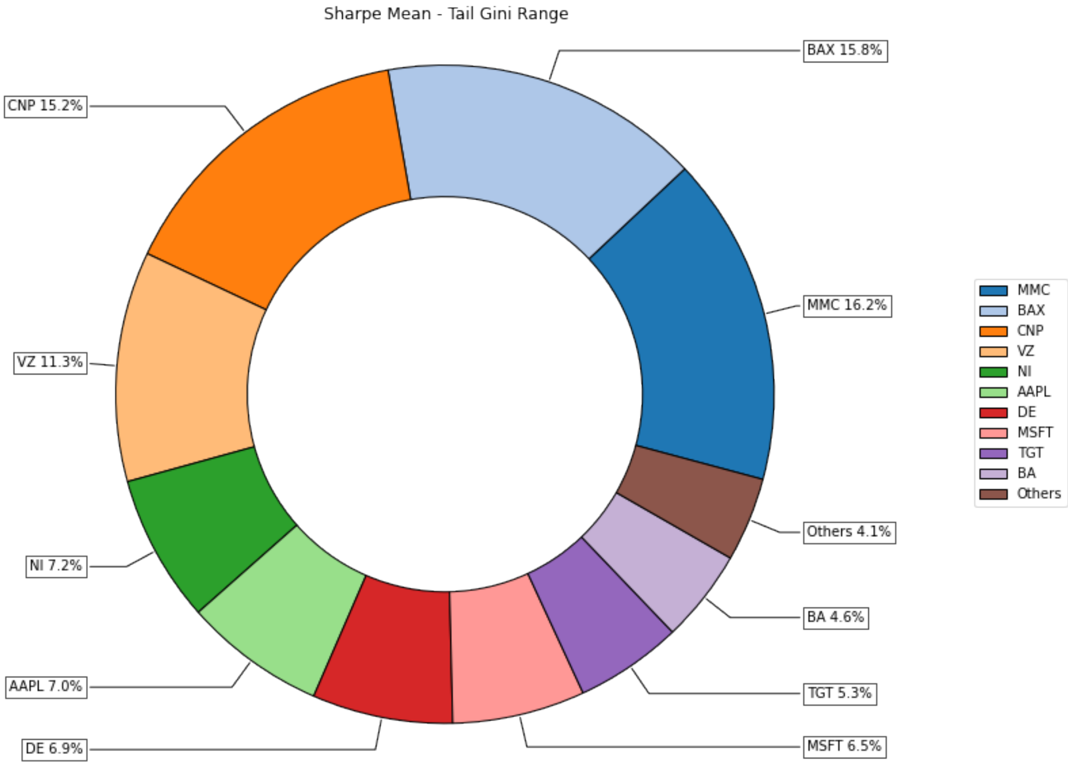

Figure 10: Composition Max Tail Gini Range Adjusted Return Portfolio

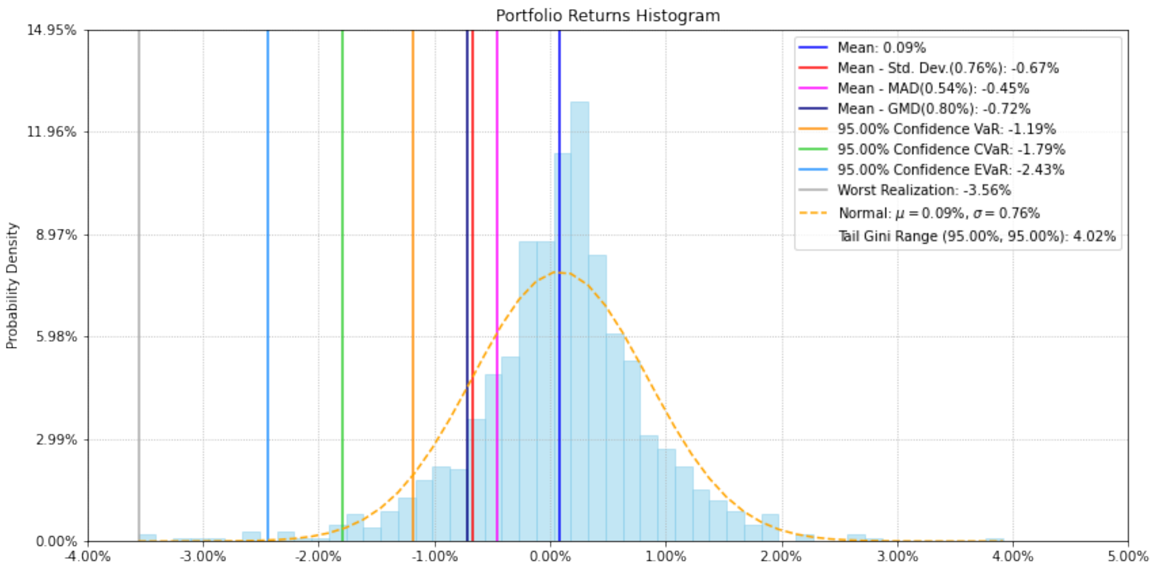

Figure 11: Histogram Risk Parity Tail Gini Range Portfolio

If we see the histogram of max tail Gini range adjusted return portfolio in Figure 11, the tail Gini range is 4.02%.

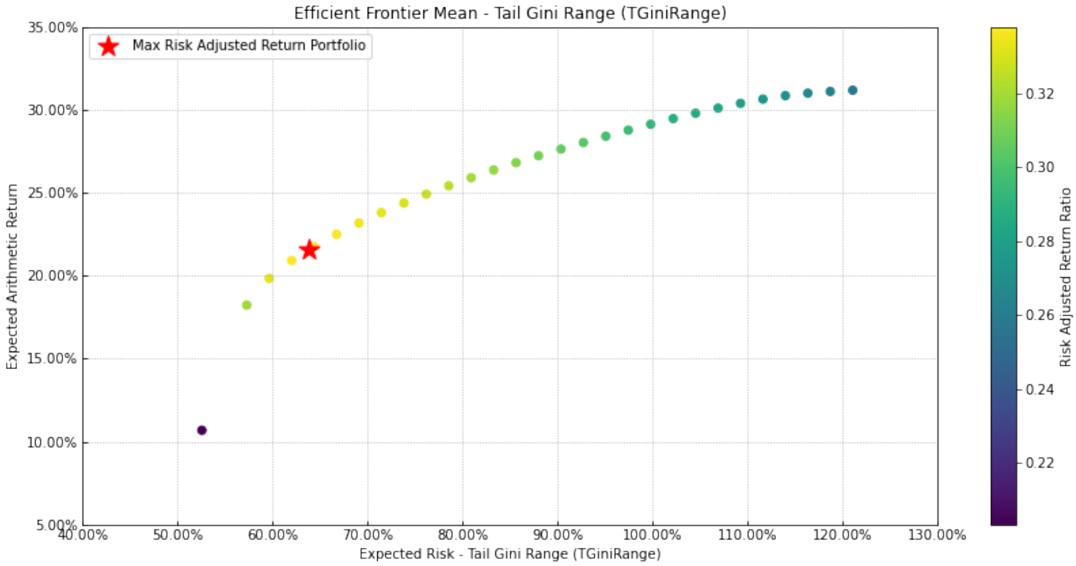

Figure 12: Efficient Frontier Return-Tail Gini Range

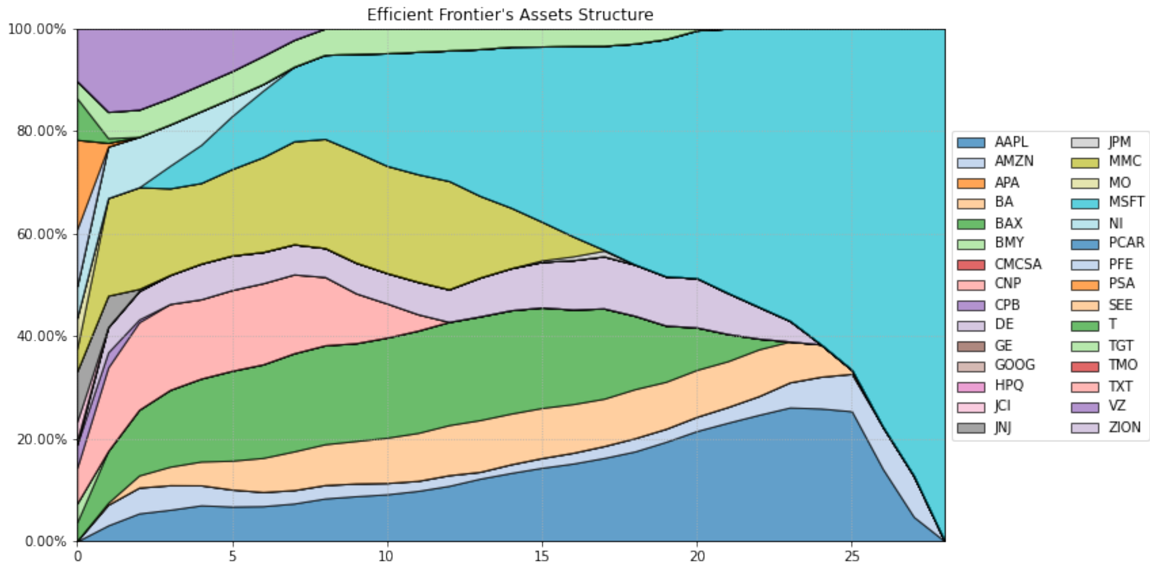

Figure 13: Composition Efficient Frontier Return-Tail Gini Range

To build the efficient frontier Return-Tail Gini Range we solve the problem maximize the portfolio return subject to a constraint on tail Gini range for several values of tail Gini range.

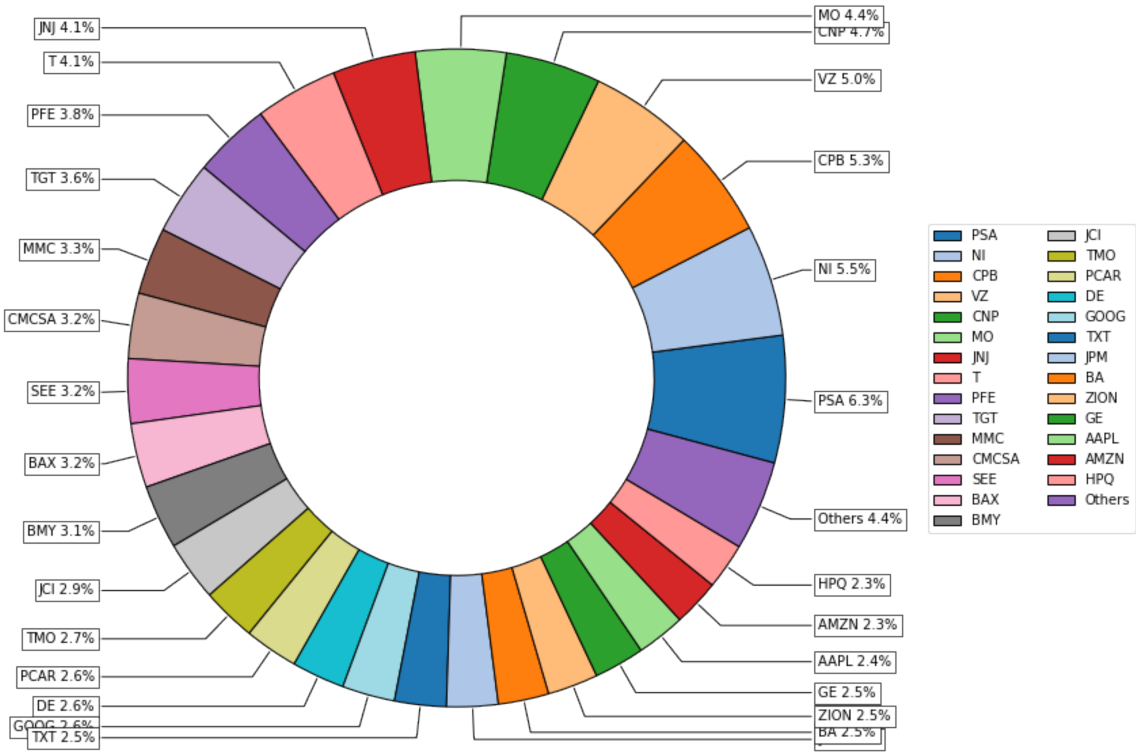

Figure 14: Composition Risk Parity Tail Gini Range Portfolio

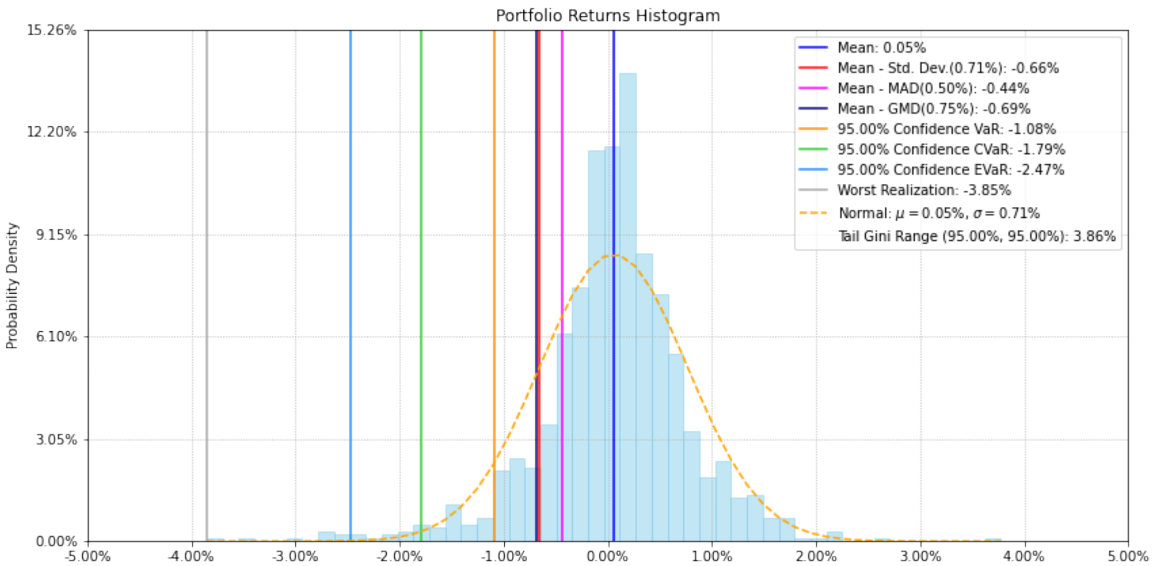

Figure 15: Histogram Risk Parity Tail Gini Range Portfolio

In the histogram of risk parity tail Gini range portfolio in Figure 14, the tail Gini range is 3.86%, this value is lower than the observed in 11. This behavior has sense because tail Gini range is a dispersion risk measure and the risk parity approach tries to make more stable the portfolio returns.

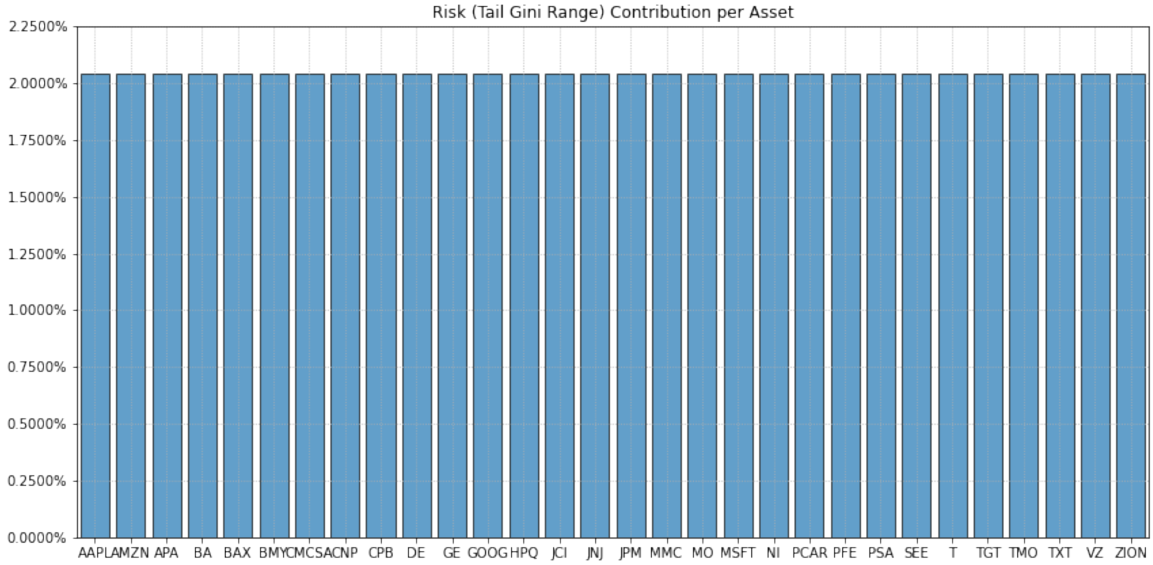

Figure 16: Tail Gini Range Contribution per Asset

In Figure 16, we can see that risk parity tail Gini range portfolio effectively breaks down tail Gini range evenly across all assets. This shows that the solution of risk parity tail Gini range problem that we posed fulfills its objective.

6 OWA Portfolio in Medium Scale Problems

In this section we compare the performance of OWA portfolio models respect to classical portfolio optimization models. We simulated 500, 700 and 1,000 normal returns for 10, 50, 100 and 200 assets. Then, we estimated 3 instances of the portfolios that maximizes the risk adjusted return and minimize risk for each risk measure and sample size. Then, we computed the average time for each combination of sample size, objective function and risk measure. The simulation was computed using a computer with four 3.6-GHz processors and 8 GB of RAM, Python 3.7, CVXPY and MOSEK solver.

Objective		Min Risk			Return/Risk		
Risk Measure	Assets	Classical	OWA	Δ	Classical	OWA	Δ
GMD	10	8.06	7.08	0.98	8.04	7.93	0.11
	50	36.89	8.10	28.79	34.51	7.62	26.89
	100	-	9.86	-	-	8.80	-
	200	-	8.27	-	-	10.50	-
CVaR	10	0.05	3.83	-3.78	0.05	3.58	-3.53
	50	0.09	3.49	-3.4	0.08	3.67	-3.59
	100	0.15	3.88	-3.73	0.14	4.44	-4.3
	200	0.32	5.18	-4.86	0.25	4.81	-4.56
WCVaR	10	0.11	4.16	-4.05	0.09	3.92	-3.83
	50	0.21	3.87	-3.66	0.21	3.74	-3.53
	100	0.41	4.38	-3.97	0.39	5.17	-4.78
	200	0.85	5.63	-4.78	0.88	4.70	-3.82
Tail Gini	10	4.36	4.90	-0.54	4.46	4.44	0.02
	50	13.45	4.35	9.1	14.02	3.86	10.16
	100	24.47	4.35	20.12	26.69	5.84	20.85
	200	50.34	5.89	44.45	50.39	4.49	45.9
Minimax	10	0.03	11.84	-11.81	0.04	3.41	-3.37
	50	0.08	3.53	-3.45	0.07	3.74	-3.67
	100	0.13	4.28	-4.15	0.13	4.60	-4.47
	200	0.26	4.75	-4.49	0.24	5.87	-5.63

Table 1: Average run time in seconds of classical and OWA portfolios for sample size of 500 returns

Objective		Min Risk			Return/Risk		
Risk Measure	Assets	Classical	OWA	Δ	Classical	OWA	Δ
GMD	10	16.71	16.17	0.54	16.89	14.35	2.54
	50	82.77	16.65	66.12	81.97	19.14	62.83
	100	-	20.30	-	-	20.18	-
	200	-	16.50	-	-	18.96	-
CVaR	10	0.05	8.30	-8.25	0.05	7.98	-7.93
	50	0.12	8.46	-8.34	0.10	9.33	-9.23
	100	0.20	9.11	-8.91	0.19	10.12	-9.93
	200	0.40	9.91	-9.51	0.38	10.17	-9.79
WCVaR	10	0.12	9.84	-9.72	0.11	9.72	-9.61
	50	0.34	9.19	-8.85	0.30	9.15	-8.85
	100	0.58	9.27	-8.69	0.55	8.63	-8.08
	200	1.21	10.72	-9.51	1.17	9.97	-8.8
Tail Gini	10	6.18	15.05	-8.87	6.90	17.43	-10.53
	50	18.21	10.04	8.17	19.76	9.55	10.21
	100	30.80	10.17	20.63	33.14	10.45	22.69
	200	67.15	13.42	53.73	65.46	9.80	55.66
Minimax	10	0.04	7.56	-7.52	0.04	10.24	-10.2
	50	0.09	8.41	-8.32	0.09	8.71	-8.62
	100	0.18	8.79	-8.61	0.17	9.17	-9
	200	0.35	9.40	-9.05	0.35	10.66	-10.31

Table 2: Average run time in seconds of Classical and OWA portfolios for sample size of 700 returns

Objective		Min Risk			Return/Risk		
Risk Measure	Assets	Classical	OWA	Δ	Classical	OWA	Δ
GMD	10	-	29.50	-	-	29.89	-
	50	-	29.96	-	-	44.02	-
	100	-	41.47	-	-	45.35	-
	200	-	32.72	-	-	53.20	-
CVaR	10	0.07	23.05	-22.98	0.06	18.95	-18.89
	50	0.16	20.60	-20.44	0.14	21.02	-20.88
	100	0.30	19.26	-18.96	0.26	22.44	-22.18
	200	0.64	19.55	-18.91	0.58	26.58	-26
WCVaR	10	0.15	25.50	-25.35	0.13	19.58	-19.45
	50	0.47	20.80	-20.33	0.40	21.37	-20.97
	100	0.93	20.12	-19.19	0.77	22.53	-21.76
	200	1.68	23.36	-21.68	1.60	20.23	-18.63
Tail Gini	10	9.23	25.63	-16.4	8.68	23.58	-14.9
	50	26.08	26.71	-0.63	28.03	30.38	-2.35
	100	49.08	28.42	20.66	50.03	28.60	21.43
	200	122.93	25.61	97.32	100.70	25.67	75.03
Minimax	10	0.06	17.88	-17.82	0.05	23.25	-23.2
	50	0.13	32.96	-32.83	0.13	18.70	-18.57
	100	0.22	20.60	-20.38	0.21	19.14	-18.93
	200	0.51	28.87	-28.36	0.44	24.29	-23.85

Table 3: Average run time in seconds of Classical and OWA portfolios for sample size of 1,000 returns

We can see in Table 1, 2 and 3 that average run time of OWA portfolios in the case of GMD and Tail Gini risk measures are lower than classical models for larger number of assets (more or equal than 100). In the case of GMD the classical models consume all RAM and freeze the computer without returning a result after several hours when the number of assets is 100 and 200, and also when the sample size is 1,000. In the case of Tail Gini, we are approximating this risk measure as a WCVaR of 100 CVaRs.

In the case of the CVaR, WCVaR and Minimax, the classical models are more efficient than OWA portfolios in average runtime independently of sample size and number of assets. We have to mention that to calculate the WCVaR, we used significance levels of 5.0%, 2.5% and 1.0% and weights of 45%, 35% and 20%.

		Objective	
Risk Measure	Assets	Min Risk	Return/Risk
CVaR Range	10	4.43	4.36
	50	4.15	4.11
	100	4.19	4.71
	200	5.31	5.82
WCVaR Range	10	7.62	4.16
	50	4.16	5.42
	100	4.27	4.27
	200	4.82	5.73
Tail Gini Range	10	5.08	5.53
	50	4.75	5.50
	100	5.18	5.67
	200	5.46	6.63
Range	10	3.55	3.62
	50	3.84	4.06
	100	4.15	4.75
	200	4.58	6.10

Table 4: Average run time in seconds of range risk measures portfolios for sample size of 500 returns

		Objective	
Risk Measure	Assets	Min Risk	Return/Risk
CVaR Range	10	13.20	9.63
	50	9.78	8.71
	100	9.81	11.11
	200	9.18	11.32
WCVaR Range	10	22.20	9.04
	50	10.73	11.10
	100	10.35	11.03
	200	10.67	11.20
Tail Gini Range	10	18.13	10.40
	50	13.22	11.04
	100	10.66	12.34
	200	11.20	11.23
Range	10	6.61	22.88
	50	8.58	9.08
	100	8.31	10.25
	200	9.26	10.97

Table 5: Average run time in seconds of range risk measures portfolios for sample size of 700 returns

		Objective	
Risk Measure	Assets	Min Risk	Return/Risk
CVaR Range	10	26.72	54.42
	50	23.17	21.63
	100	26.14	25.30
	200	23.03	28.56
WCVaR Range	10	42.62	23.72
	50	24.96	24.93
	100	21.93	22.50
	200	23.31	28.46
Tail Gini Range	10	29.26	27.81
	50	27.50	27.21
	100	28.57	28.97
	200	30.30	27.05
Range	10	17.78	20.86
	50	19.14	20.67
	100	18.90	23.63
	200	20.00	26.85

Table 6: Average run time in seconds of range risk measures portfolios for sample size of 1,000 returns

We can see in Table 4, 5 and 6 that range risk measures have average runtime very similar to those observed for their downside risk measure version. On the other hand, we can not compare OWA portfolios using range risk measures with classical model versions, because there are no models in academic literature that study this risk measures.

Finally, we can notice that in contrast to classical models, the average runtime of OWA portfolios are less sensitive to the number of assets, this model is more sensitive to the sample size.

7 Conclusions

The OWA portfolio optimization model is a general and flexible framework that allows us to model several classic risk measures and to define a new class of risk measures called range risk measures. Also, we can define personalized risk measures depending on the weights of the OWA operator.

This work presents a disciplined convex programming framework for OWA portfolio optimization, that allows us to propose five OWA portfolio optimization problems. Then, we make some examples using Python, CVXPY and MOSEK solver. We built an efficient frontier, a maximum risk adjusted return portfolio, and a risk parity portfolio using GMD, tail Gini and Range risk measures. Also, we performed a simulation to test the efficiency of the proposed frameworks and we observed that the new OWA portfolio framework in the case of GMD and tail Gini has a better performance than the models proposed by Yitzhaki (1982) and Ogryczak and Ruszczynskia (2002) respectively.

The models proposed in this work allows users to use disciplined convex programming softwares that supports linear programming to calculate OWA portfolio optimization models like GMD, tail Gini and range risk measures. The case of GMD is important because the OWA version substantially reduces the time and resources used to solve the GMD portfolio optimization model. This formulation makes portfolio optimization based on OWA risk measures accessible to most users.

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